xT Pharmacokinetic Model

Validation Dossier: supported medications, model derivation, accuracy assessment, and known limitations

Engineering validation pass

31 July – 1 August 2026

Abstract

This document describes every medication and route xT models, how each prediction is derived, what published data each constant is anchored to, and how accurate the resulting predictions are. It is written to be checkable: every figure quoted is regenerated by a script in `.pk_analysis/`, and the provenance of every constant is stated, including where that provenance is weak.

The headline results of this validation pass are: two constant-lookup defects that between them overpredicted by roughly 1,090×, 5,300× and 344× on three medication/route combinations, plus a third defect in the chart's forward projection; a testosterone cypionate recalibration against a primary population-PK source; a conservation-of-mass correction that brought the three remaining testosterone esters into agreement with it; and a set of structural limitations that are quantified here rather than fixed, because fixing them requires a different model rather than different numbers.

Two of those defects were found by running synthetic user scenarios (Section 7) rather than by inspecting constants, which is the main methodological conclusion of the exercise.

Sections 8 and 9 are the ones to read if you are deciding whether to trust a particular number. Nothing in this document should be read as clinical guidance.

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1 Scope: what the app models

xT predicts serum hormone concentration over time from a user's logged dose history. Two medication classes are supported, each across three administration routes.

Medication	Route	Ester / form	Output unit
Estradiol	Injectable	Valerate (EV), Cypionate (EC)	pg/mL
Estradiol	Oral	(route pseudo-ester)	pg/mL
Estradiol	Cream	(route pseudo-ester)	pg/mL
Testosterone	Injectable	Cypionate (TC), Enanthate (TE), Propionate (TP), Undecanoate (TU)	ng/dL
Testosterone	Oral	undecanoate (route pseudo-ester)	ng/dL
Testosterone	Cream	unesterified gel (route pseudo-ester)	ng/dL

Table 1: The ten medication/route/form combinations the model covers.

Route pseudo-esters. The two non-injectable cases in each ester enumeration describe a *route*, not a chemical ester. They are resolved from the dose record's administration method by `MedicationManager.effectiveEsterName`, never chosen by the user. That resolution was introduced in this pass and is the fix for the largest defect described in Section 7.

What is modeled separately. The engine returns the *exogenous* contribution only. Endogenous production is added by the app as a separate `baseLevel` term that decays as gonadal suppression progresses. This matters for interpreting several constants below, most importantly testosterone cypionate's half-life.

2 The model

2.1 Governing equation

Each administered dose contributes a Bateman biexponential — first-order absorption from a depot into a single well-mixed compartment, first-order elimination from it:

$$C_i(t) = D_i \cdot \mathrm{EDF} \cdot S \cdot \frac{k_a}{k_a - k_e}\left(e^{-k_e t} - e^{-k_a t}\right), \qquad t \geq 0 \tag{1}$$

where D_i is the administered dose in mg, EDF is the effective dose factor (Section 2.4), S is a scaling constant with units of concentration per mg, and k_a, k_e are rate constants in day^{-1}. Total predicted level at time t is the superposition of all doses taken before it, plus the endogenous baseline, times a user calibration factor:

$$C(t) = \alpha \left[C_{\mathrm{base}}(t) + \sum_{i\,:\,t_i \leq t} C_i(t - t_i) \right] \tag{2}$$

Superposition is exact here because Equation 1 is linear in dose. That linearity is also the model's principal simplification: there is no saturable protein binding, no SHBG dynamics, no dose-dependent clearance.

The $k_a \approx k_e$ limit. When $|k_a - k_e| < 10^{-4}$ Equation 1 is singular, and the engine substitutes its limit, $D \cdot \mathrm{EDF} \cdot S \cdot k_a\, t\, e^{-k_a t}$.

Transdermal. The cream/gel route does not use Equation 1. It is modeled as 24 hourly microdoses of $D/24$ each, decaying single-exponentially, approximating a 24-hour square-wave release. Section 8 explains why this under-models intraday variation.

2.2 Time to peak

Differentiating Equation 1 gives

$$t_{\max} = \frac{\ln(k_a/k_e)}{k_a - k_e} \tag{3}$$

Note that Equation 3 is *symmetric* under $k_a \leftrightarrow k_e$. This matters more than it looks: for depot esters the two exponentials are not separately identifiable from clinical data, so which one is labelled "absorption" and which "elimination" is a modelling convention rather than a finding. The observable curve is unchanged by the swap, with the amplitude difference absorbed into S.

2.3 Flip-flop kinetics

For depot esters, absorption from the oil vehicle is slower than elimination of the released hormone ($k_a \ll k_e$). The terminal slope of the curve is then governed by *absorption*, not elimination — "flip-flop" kinetics. The practical consequence is that the field named `halfLifeDays` in the code sets k_a and controls the entire inter-dose decay, while `eliminationHalfLifeHours` sets k_e, which mostly controls where the peak falls. The names are historical and slightly misleading; the doc comments say so.

2.4 Effective dose factor

EDF = EWF × F converts administered mg of ester into systemically available mg of active hormone.

The *ester weight factor* is pure stoichiometry — the mass fraction of the ester molecule that is the parent hormone:

$$\text{EWF} = \frac{M_{\text{hormone}}}{M_{\text{ester}}}$$

with $M_{\text{E}_2} = 272.4$ and $M_{\text{T}} = 288.4$ g/mol. These are the only constants in the entire model that are not fitted to anything and cannot be wrong.

F is route bioavailability. Injectable depot $F = 1.0$; oral estradiol $F = 0.05$ (first-pass metabolism); transdermal estradiol $F = 0.15$; transdermal testosterone $F = 0.10$, which the AndroGel 1% label states outright in §12.3. Oral testosterone undecanoate carries $F = 1.0$ not because it is fully bioavailable — it is not — but because its absolute bioavailability is not established in the source. Since F and S are confounded (only their product is identified by the published data), folding it into S avoids inventing a number. Predictions are unaffected by where the split is drawn.

2.5 The scaling constant

S is an empirical surrogate for F/V_d. It is the one constant with no direct physical interpretation, and historically the one most often wrong. This pass replaced curve-fitting for S with closed forms wherever the sources allowed; see Sections 4.2 and 4.3.

3 Encoded constants

Reading Table 2. Three things in it are worth noticing before the derivations.

	slow exponential		fast				
Form	$t_{1/2}$ (d)	k_a (d^{-1})	k_e (d^{-1})	EWF	F	S	t_{max} (d)
EV	2.6000	0.2666	0.7737	0.764	1.00	396.40	2.10
EC	6.4700	0.1071	0.4310	0.687	1.00	273.00	4.30
E2 oral	0.0791	8.7629	1.3859	1.000	0.05	1250.00	0.25
E2 cream	1.5000	0.4621	5.5452	1.000	0.15	155.00	—
TC	3.8390	0.1806	1.2196	0.699	1.00	67.13	1.84
TE	7.0000	0.0990	8.3178	0.720	1.00	457.67	0.54
TP	1.0000	0.6931	8.3178	0.837	1.00	457.67	0.33
TU	20.0000	0.0347	8.3178	0.631	1.00	457.67	0.66
T oral (TU)	0.0900	7.7016	8.3053	0.631	1.00	13.57	0.13
T gel	0.5784	1.1984	—	1.000	0.10	132.98	—

Table 2: Constants as shipped, and the t_{max} each implies via Equation 3. The cream routes have no t_{max} because they are not modeled with Equation 1; E2 cream's k_e is stored but unused.

TC's slow half-life (3.84 d) is now *shorter* than TE's (7.0 d), inverting the previous ordering even though the two esters are clinically near interchangeable. This is an artifact of TC being the only ester with a primary-source refit, not a claim about relative clearance.

TE, TP and TU share an identical S. That is not a copy-paste error; it follows from the identity in Section 4.3 and the fact that all three still share a k_e.

E2 oral's stored half-lives were first *restated* (0.5 d → 0.2773 d, 3.0 h → 12.0035 h) without changing any prediction. The engine had never read those fields for oral doses; it hardcoded $k_a = 2.5$ and $k_e = 1.386$ inline, and $S = 1250$ was fit against those. Restating them made the constants stop disagreeing with the code — and, once they were visible, made it possible to see that k_a was wrong.

It has since been *refit*: 0.2773 d → 0.0791 d, i.e. $k_a = 2.5 \to 8.762\,d^{-1}$, moving t_{max} from 12.7 h into the published window at 6.0 h. Unlike the restatement, this one moves predictions. k_e and S are unchanged; the reasoning for holding S is in Section 8.

4 Derivations by form

4.1 Estradiol valerate and cypionate

Recalibrated on 29 July 2026 (before this pass) against transfemscience.org's compiled tables, which aggregate published single-dose and steady-state estradiol PK. k_e was fit so Equation 3 reproduces the published time-to-peak; k_a was fit so the modeled steady-state trough reproduces Table 10's 5 mg-weekly figures; S was anchored to the same table's steady-state peak.

The important decision recorded there, and re-made in this pass for testosterone, is that single-dose and steady-state targets were **not simultaneously satisfiable**, and steady state was chosen. Section 9 returns to this.

4.2 Testosterone cypionate

TC is the only form in the model derived from a primary population-PK publication: Bi et al. (2018), *CPT: Pharmacometrics & Systems Pharmacology*, `doi:10.1002/psp4.12287`. Table 2 of that paper reports $CL/F = 2.6$ kL/day, $V/F = 14.4$ kL, and $K_a = 1.22\,day^{-1}$.

Rate constants. The disposition rate is $CL/V = 2.6/14.4 = 0.1806\,day^{-1}$, giving a half-life of 3.839 d, which the code stores as `halfLifeDays`. $K_a = 1.22$ gives $\ln 2/1.22 = 13.64$ h, stored as `eliminationHalfLifeHours`. (Per the symmetry noted after Equation 3, the labels are conventional.)

Why 3.84 days and not 7–8. The commonly quoted TC half-life is around a week, and Bi addresses the discrepancy directly. Noncompartmental analysis of their own data yields 6.9 d, and their model yields 6.87 d *when endogenous secretion is assumed to be zero* — but 4.05 d once endogenous production is accounted for. Because xT models endogenous production separately as `baseLevel`, the exogenous-only figure is the one that belongs in the ester constant. Bi's own median post-hoc estimate of 4.05 d agrees with our 3.839 d to 5.2%.

Scaling constant, in closed form. Matching Equation 1 to the one-compartment model underlying Bi's parameters gives S analytically rather than by curve fit:

$$S = \frac{10^5 \cdot K_a}{V_L \cdot \mathrm{EDF} \cdot k_e} = \frac{10^5 \cdot 1.22}{14400 \times 0.699 \times 0.180556} = 67.13 \tag{4}$$

The 10^5 converts mg/L to ng/dL; EDF appears because the app multiplies the dose by it while Bi's V/F already absorbs ester conversion. This reproduces Bi's population model exactly, which is why the corresponding row of Table 4 shows 0.0% error — that row is a consistency check, not evidence.

Resolution of a standing concern. The project's own `OPEN_CONCERNS.md` §2 had flagged the encoded TC $t_{\max}$ window (1–2 d) as probably wrong, because a secondary source placed $C_{\max}$ on day 4–5. The primary source settles it in favour of the encoded window: Bi's parameters give $t_{\max} = 1.84$ d via Equation 3, and propagating the published 95% CI on K_a (0.785–2.67) gives a band of 1.08–2.43 d, which never reaches day 4–5. Read TC $t_{\max}$ as 1.84 ± 0.7 d; Bi's own limitations section (1–2 absorption-phase samples per subject, K_a RSE 44.9%) explains the imprecision.

4.3 Testosterone enanthate, propionate, undecanoate

None of these three has a primary source in hand. Their scaling constants are nonetheless *not* free parameters, because of a conservation argument that follows from the TC refit.

Integrating Equation 1 over all time gives exactly

$$\int_0^\infty \frac{k_a}{k_a - k_e}\left(e^{-k_e t} - e^{-k_a t}\right)\, dt = \frac{k_a}{k_a - k_e}\left(\frac{1}{k_e} - \frac{1}{k_a}\right) = \frac{1}{k_e}$$

so at steady state with dosing interval τ, superposition gives

$$C_{\mathrm{avg}} = \frac{D \cdot \mathrm{EDF} \cdot S}{\tau\, k_e} \tag{5}$$

The true steady-state average is (mg of testosterone per day) divided by clearance — and clearance is a property of *testosterone*, not of the ester carrying it. It is therefore already pinned by the TC refit: $\mathrm{CL}_T = 2600 \times \mathrm{EWF}_{TC} = 1817.4$ L/day. Equating the two expressions, the ester weight factor cancels:

$$\boxed{S = \frac{10^5 \cdot k_e}{\mathrm{CL}_T \cdot F}} \tag{6}$$

S depends only on k_e and clearance. The ester's half-life — k_a — does not appear at all, which is worth stating plainly because two now-deleted unit tests asserted an ordering of S by half-life and were reasoning from the wrong quantity.

The load-bearing check. TC satisfies Equation 6 to 0.03% *without having been fit to it* — its S came from Equation 4, via V/F and K_a. Two independent derivations landing on the same number is what makes it defensible to apply the identity to the other three.

Ester	S before	S now	S implied	error before	**Regimen**	C_{avg} before	after
Cypionate	1081.0	67.13	67.11	16.11×	100 mg q7d	1298	550
Enanthate	923.0	457.67	457.67	2.02×	100 mg q7d	1141	566
Propionate	359.0	457.67	457.67	0.78×	50 mg q2d	903	1151
Undecanoate	214.0	457.67	457.67	0.47×	1000 mg q70d	232	496

Table 3: The clearance identity applied. C_{avg} in ng/dL, exogenous only. The "before" column for cypionate also reflects its k_e moving from 2.0 h to 13.64 h.

What was wrong. The previous values spanned 0.47× to 2.02× of the implied value and were *mutually inconsistent* — the app told users on different esters different things about the same clearance. Enanthate is the clearest case: a standard 100 mg weekly regimen predicted $C_{\max} = 1753$ ng/dL including baseline, a level that reads as clearly supraphysiological, while contradicting its primary-sourced sibling by a factor of two.

What this does not fix. Magnitude only. TE, TP and TU retain a placeholder 2 h k_e, so their $t_{\max}$ (Table 6) is badly wrong. See Section 8.

4.4 Transdermal testosterone

Anchored to the AndroGel 1% prescribing information, Table 5 (Day 180 steady state, $n = 44$): 50 mg/day → $C_{\text{avg}} = 555$ ng/dL. The terminal half-life comes from the label's washout statement — levels return to baseline in 48–72 h, implying ∼5% residual at 60 h, giving $t_{1/2} = 0.5784$ d.

Two limitations were documented rather than fitted around; both are in Section 8.

4.5 Oral testosterone undecanoate

Anchored to Swerdloff et al. (2020) / the Jatenzo label: 237 mg twice daily → $C_{\text{avg}} = 488.7$ ng/dL, $t_{\max}$ 2–4 h. Both rate constants were solved jointly so that $t_{\max}$ lands at 3 h and the steady-state $C_{\max}/C_{\text{avg}}$ comes as close as the model permits to the reported 2.06. Section 8 explains why it cannot reach it.

5 Accuracy

5.1 Against the anchors

Regimen	**Quantity**	**Model**	**Published**	**Error**	**Source**
EV 5 mg q7d	$C_{\max}$	384	384	+0.1%	TFS Table 10
EV 5 mg q7d	$C_{\min}$	142	142	+0.1%	TFS Table 10
EC 5 mg q7d	$C_{\max}$	339	339	−0.1%	TFS Table 10
EC 5 mg q7d	$C_{\min}$	262	262	−0.1%	TFS Table 10
TC 100 mg q7d	C_{avg}	550	549	+0.0%	Bi Table 2
T gel 50 mg/d	C_{avg}	555	555	+0.1%	AndroGel Table 5
T gel 75 mg/d	C_{avg}	833	601	+38.6%	AndroGel Table 5
T gel 100 mg/d	C_{avg}	1111	713	+55.8%	AndroGel Table 5
Oral TU 237 mg BID	C_{avg}	489	489	−0.0%	Jatenzo
Oral TU 237 mg BID	$C_{\max}$	789	1008	−21.7%	Jatenzo

Table 4: Predicted versus published. Rows with near-zero error are the anchors themselves and demonstrate only that the fit converged. The three coloured rows are the informative ones; both are explained in Section 8. Estradiol in pg/mL, testosterone in ng/dL.

5.2 Out-of-sample checks

Rows in Table 4 with 0.0% error prove nothing — a fit agrees with its own target by construction. The following comparisons were *not* used in any fit.

Check	Model	Published	Error	Source
TC single 200 mg C_{max} (exogenous)	997	872	+14.3%	200 mg IM lit. 1112 less 240 basal
TC 100 mg q7d C_{max} + basal	976	1150	−15.1%	Bi Figure 2, 100 mg panel
EV single 2 mg C_{max}	119	118	+1.0%	TFS Table 9
EC single 2 mg C_{max}	59	62	−5.1%	TFS Table 9

Table 5: Independent validation. The estradiol rows are the strongest evidence in this document: constants fit purely to steady-state data reproduce single-dose peaks within 1% and 5%.

The two testosterone rows agree to about 15% in opposite directions, which is roughly the precision the source itself supports given Bi's reported RSEs. They are consistency checks, not tight validation.

5.3 Time to peak

Form	Model t_{max}	Published	Source quality
EV	2.10 d	2.1 d	compiled tables
EC	4.30 d	3.1–4.3 d	compiled tables
E2 oral	6.0 h	4–8 h	secondary
TC	1.84 d	1.84 d	**primary** (Bi Table 2)
TE	12.9 h	1.5–7 d	secondary
TP	7.8 h	∼0.6 d	sparse
TU	15.9 h	∼7 d	secondary (Wang 2010)
T oral	3.0 h	2–4 h	label

Table 6: Where the model puts the peak. This is the table that shows what is still uncalibrated: magnitude is right everywhere, shape is not.

6 Persona validation

Section-level agreement with published tables is necessary but not sufficient: those tables all describe clean steady state on a fixed regimen, and real users miss doses, change dose, change route, and dose on intervals nobody studied. Eight synthetic users were run end to end through the same summation the app performs. Full output: `python3 .pk_analysis/personas.py`.

Persona	Regimen	What it tests, and the result
Maya, 24	4 mg EV IM q7d, 18 months, perfect adherence	Control. C_{max} 307, C_{min} 114 pg/mL — exactly 4/5 of the 5 mg source figures, as linearity requires. Pass.
Devon, 31	100 mg TC q7d → 200 mg q14d at month 6	Superposition across a regimen change. C_{avg} preserved at 790 ng/dL as it must be; trough falls 561 → 381 while peak rises 976 → 1339, and peak/trough widens 1.74× → 3.51×. The model reproduces the clinical trade-off and correctly refuses to reward the longer interval. Pass.

Persona	Regimen	What it tests, and the result
Priya, 47	2 mg oral E2 BID for 4 months, then 4 mg EV q7d	Route switch; historical doses must keep their own kinetics. They do — the oral segment retains $F = 0.05$ and the fast rates. This persona also found the oral t_{max} defect, which collapsed her modeled intraday swing to 1.1×; after the k_a refit it is 1.37× at an unchanged C_{avg} of 180 pg/mL. Pass.
Sam, 19	50 mg/day T gel, misses ∼2 days/week	The configuration that used to overpredict by three orders of magnitude, plus skipped-dose handling. Now C_{avg} 637 ng/dL; his missed days cost 20% of his average and 63% of his trough. Pass.
Rowan, 38	4 mg/day compounded E2 cream, onboarded in-app	Found a defect: onboarding stored the ester as `"Cream"`, matching no enum case. Fixed at both ends.
Kai, 52	237 mg oral TU BID, 10-day interruption	Washout and re-accumulation on the fastest route. Falls to the endogenous floor within a day; no phantom drug survives the gap; recovers to 100% of pre-gap within 2 days. Pass.
Jules, 28	100 mg TE IM q7d	Found a defect: steady-state C_{avg} was 2× too high and contradicted primary-sourced TC. Fixed by Equation 6 (C_{max} 990, C_{min} 637 ng/dL). Shape still wrong — t_{max} 0.46 d.
Nadia, 35	5 mg EC q10d — an interval nobody studied	Extrapolation outside the calibration envelope. Trough holds in band but sits 40% below the same dose at q7d. Structurally sound but inherits every assumption of the fit. Pass, with caveat.

All eight land inside their clinical reference bands. Two of the eight found defects that inspection had missed, which is the argument for running scenarios rather than only checking constants.

7 Defects found and fixed

7.1 Non-injectable routes fell through to hardcoded constants

The scaling-constant lookup inside `estimatedLevels` was guarded on `medication == "Estrogen"`. A *testosterone* dose on the Oral or Cream route therefore matched no branch and fell through to the hardcoded literals `12000.0` and `8000.0` — which are the pre-28-July *estrogen* values that the codebase already documents as 9.6× and 52× too large even for estrogen.

Three errors compounded: the wrong S; the wrong EDF (resolved from the stored injectable ester rather than the route, so a gel dose used cypionate's 0.699×1.0 instead of the gel's 1.0×0.10); and a hardcoded absorption rate that also changed day-to-day accumulation.

Reachability. This was not theoretical. The settings screen offered every administration method for every medication type, while the ester list returned an empty array for testosterone + Oral/Cream — so the previously selected injectable ester stayed silently attached to the record and the user saw no ester picker at all.

Scenario	Before	After	Factor	Published
T gel 50 mg/d	605,183	555	1,090×	555
Oral TU 237 mg BID	2,589,554	489	5,299×	489
E2 cream 4 mg/d, onboarded	69,263	201	344×	—

Table 8: Steady-state C_{avg}, ng/dL for testosterone and pg/mL for estradiol. These are simulated through both code paths, not estimated from constant ratios.

Fix. `MedicationManager.effectiveEsterName(ester:route:)` makes the *route* decide which constants apply, and both `pkParams` and `dosePotencyFactor` now go through it. The hardcoded literals and rates are gone; every route reads its constants from the ester enumeration. The settings and onboarding ester lists were narrowed to `injectableCases`.

7.2 Onboarding stored an ester that matched nothing

A cream dose created through onboarding was saved with ester `"Cream"` — the raw administration-method name, which matches no case in `MedicationEster`. The same lookup failure applied, so estradiol cream users who onboarded in-app got the hardcoded 8000 while users who configured the identical route from Settings stored `"Transdermal"` and were fine.

Fixed at both ends: onboarding now stores the route pseudo-ester, and the route normalization above repairs records that already exist.

7.3 Forward projection assumed a 7-day interval for every injectable

With exactly one logged dose the chart projects a future cadence so the user can preview a trajectory. That interval came from the administration method, which returns 7 days for anything injectable. For TU — dosed every 10–14 weeks — a single logged 1000 mg dose was projected as *weekly* 1000 mg injections, reaching ∼2850 ng/dL against a single-dose ∼1190. TP, dosed every other day, was under-projected for the mirror-image reason.

Fixed by adding `typicalIntervalDays` to both ester enumerations. A user-supplied frequency still takes precedence.

8 Known limitations

These are quantified rather than fixed. Each needs either a source we do not have or a structurally different model.

8.1 TE, TP and TU have the right magnitude and the wrong shape

All three retain a placeholder 2 h k_e, which pins their $t_{\max}$ to 0.3–0.7 d against published windows of 1.5–7 d (Table 6). The peak arrives days early and the curvature between peak and trough is wrong. Steady-state average is correct.

They were not fitted because the usable TE steady-state profile is *subcutaneous* — which absorbs more slowly and flatter than the intramuscular route modeled here, so its k_a does not transfer without justification — and because that figure comes from a secondary summary rather than a primary table. TP's published data is sparse.

A visible consequence is deliberately left in place: TU's modeled $t_{\max}$ (0.66 d) is now *earlier* than TC's (1.84 d), inverting the clinical ordering. A unit test asserts this inversion and instructs whoever fixes TU's k_a to delete the test and restore the clinical assertion. That test passed before this pass only because TC was also wrong, in the same direction, by more.

8.2 Testosterone undecanoate needs a three-parameter model

Over a 70-day interval the two-parameter Bateman produces a peak/trough ratio of about 11×. The real drug is 2–3×. No choice of (k_a, k_e, S) reproduces that flatness.

The magnitude correction in Section 4.3 improves TU's C_{avg} (232 → 496 ng/dL exogenous, against a published ∼500–600 total) but makes its $C_{\max}$ overshoot, because correcting magnitude cannot correct shape. That trade was taken knowingly: C_{avg} is what clinicians titrate on and what a user lives at, and the peak was already wrong.

8.3 Transdermal intraday ripple is structurally under-modeled

The 24-hourly-micro-dose construction has no absorption phase and tops out at $C_{\max}/C_{\min} \approx 1.05$ for testosterone gel, against an observed 2.24. Reproducing 2.24 would require a terminal half-life near 1 h, which contradicts the label's own 48–72 h washout by an order of magnitude. The two cannot be satisfied together in this model. C_{avg} is exact.

8.4 Transdermal dose-response is anchored to one arm

The AndroGel 1% label's dose-response is markedly sublinear: 50 mg → 555, 75 mg → 601, 100 mg → 713 ng/dL. The 1.62% label inverts outright — 81 mg C_{avg} 537 is *below* 60.75 mg C_{avg} 643.

This is a titration artifact: higher-dose arms are enriched for poor absorbers, because subjects were escalated precisely when they responded badly. It is selection confounding, not saturable pharmacokinetics, and a linear model should not be bent to reproduce it. S is anchored to the 50 mg arm — the standard starting dose and the least confounded — so predictions at 75 and 100 mg run high by 39% and 56% (Table 4).

Gel absorption also varies enormously between individuals. The app's calibration factor against the user's own blood tests is the intended remedy for this and is more important on this route than on any other.

8.5 Oral testosterone under-predicts peak fluctuation

With $t_{\max}$ pinned to the reported 2–4 h window, the model's achievable $C_{\max}/C_{\text{avg}}$ tops out at 1.61 against a reported 2.06, so $C_{\max}$ reads 789 rather than 1008 ng/dL.

The reported ratio is itself inflated by mean-of-maxima bias: the published $C_{\max}$ is the mean of per-subject maxima, and $\mathbb{E}[\max] \geq \max[\mathbb{E}]$, with a reported SD of ±581 on a mean of 1008. Anchoring to C_{avg} — the label's own primary endpoint — and accepting a documented under-prediction of the peak was judged the better trade.

8.6 Oral estradiol peaked too late — refit

$k_a = 2.5$ and $k_e = 1.386$ gave $t_{\max} = 12.7$ h against a published oral estradiol $t_{\max}$ on the order of 4–8 h. On a twice-daily schedule this mattered more than the number suggests: $t_{\max}$ exceeded the 12 h dosing interval, so each dose was still rising when the next arrived and the modeled curve flattened to a peak/trough of ∼1.1× across the day. Real oral estradiol swings considerably more.

k_a is now 8.762 /day, placing $t_{\max}$ at 6.0 h and widening the modeled BID swing to ∼1.37×. Two things in that are choices rather than fits, and are recorded as such.

First, *where* inside the window. Nothing picks a point in 4–8 h — it is a range from secondary sources, not a distribution with a mode — so $t_{\max}$ is pinned to the midpoint. Note also that Equation 3 is symmetric under $k_a \leftrightarrow k_e$, so inverting it for k_a has two roots; the fit is bracketed on the $k_a > k_e$ branch.

Second, *what to preserve*. Integrating the Bateman term gives AUC/dose $= D^{\text{eff}} S/k_e$, with no k_a in it: S and k_a are coupled through the peak but not through the average. One (S, k_a) pair therefore cannot hold both the published $t_{\max}$ and the published $C_{\max}$, which makes "refit k_a and re-derive S" a hidden choice of anchor rather than a mechanical step. We anchored on C_{avg}: 2 mg

daily stays at 90 pg/mL, unchanged, and it is the quantity clinicians titrate on. The accepted cost is single-dose C_{max} rising from 60 to 88 pg/mL for 2 mg. Holding C_{max} at 60 instead would have required $S = 848.5$ and dropped C_{avg} to 61 pg/mL — moving every average and trough figure in the app to buy back one single-dose peak. This is the same trade, and the same direction, as the oral TU decision above and the injectable re-anchor.

8.7 Model-wide simplifications

Dose-linearity with no saturable binding or SHBG dynamics; a single well-mixed compartment; constant clearance; no covariates (body composition, age, hepatic function, concomitant medications beyond the one spironolactone adjustment); and no representation of between-subject variability, which for the transdermal route in particular is the dominant source of error. Predictions are population means, not individual forecasts. The per-user calibration factor exists to close some of that gap once blood tests are available.

9 Open questions requiring non-engineering judgement

Three things in this model are decisions rather than derivations, and were made by engineering because they blocked the work. They deserve explicit ratification.

9.1 Single-dose targets versus steady-state targets

For both estradiol esters and all four testosterone esters, the single-dose peak targets written into the code's own documentation and the published steady-state data were **mutually unsatisfiable**. For estradiol the gap was large: holding the app's documented single-dose targets put steady state +110% (EV) and +206% (EC) above the published table.

Steady state was chosen, both times, on the reasoning that it is what users actually live at and what clinicians titrate on. The consequence is that the single-dose figures in the doc comments are now *derived* values, not enforced targets. If the product intent was the reverse, this needs revisiting.

9.2 Provenance of the original targets

The estradiol single-dose targets claimed provenance from Düsterberg et al. and Tiong et al., but do not match those sources (2.1× for EV, 3.1× for EC). Whoever set them either used a different reference population or assay, or made an error. That question is unresolved.

Relatedly: the previously shipped EC constant ($S = 2700$) produced 51 pg/mL for a 2 mg dose, not the 180 its own comment claimed — a −71.5% mismatch. Every other ester matched its stated target to within 0.1%, which suggests that one value was never checked against the target written beside it. Worth understanding how that survived review, since the same process produced the others.

9.3 What the TU target range is describing

The TU target range written in the code (400–700 ng/dL) sits well below the ∼1100 ng/dL C_{max} reported for a single 1000 mg injection. It may be describing an intended steady-state *trough* rather than a single-dose peak. No numeric change was made pending confirmation of which quantity it means.

9.4 Communicating the change

Predictions move substantially for most configurations in this pass — see `OPEN_CONCERNS.md` §5 for the per-form table. Nothing about anyone's actual hormone levels has changed; the previous

numbers were wrong.

No user-facing release note was written, because the app is not yet deployed and there are no existing users to inform. That decision should be revisited before launch. If any of these numbers ship and then move again, the tone will matter: users of this app are tracking a medical transition, and an unexplained apparent halving of their levels is alarming in a way that a normal changelog entry does not address.

10 Verification and reproducibility

10.1 Guards in the codebase

The Swift test suite carries the calibration claims as executable assertions. The load-bearing ones added or rewritten in this pass:

- `testosteroneScalingSatisfiesClearanceIdentity` — checks Equation 6 for every injectable T ester. Fails on drift in any of S, k_e, F or the clearance constant. This is the assertion that would have caught the TE defect.
- `cypionateScalingMatchesAnalyticForm` — checks Equation 4 independently, which is what makes the identity trustworthy.
- `cypionateTPeakTime` — pins TC $t_{\max}$ to Bi's 1.84 d and to the CI band implied by K_a's published interval.
- `undecanoatePeakTimeIsUncalibrated` — asserts the *known-wrong* TU/TC ordering, so the gap stays visible instead of being quietly re-broken.
- `testosteroneEliminationHalfLife` — asserts that TE, TP and TU still carry the 2 h placeholder, so nobody assumes they are calibrated.

10.2 Scripts

Script	Purpose
`verify_constants.py`	Independent check of the *shipped* constants. Run first.
`personas.py`	The eight end-to-end user scenarios.
`fit_testosterone.py`	Derivation of TC, gel and oral T constants.
`fit_constants.py`	Derivation of the estradiol ester constants.
`dossier_tables.py`	Regenerates every numeric table in this document.
`pk_engine_replica.py`	Python mirror of the Swift engine; all of the above import it.

`verify_constants.py` is deliberately *not* the fitting code. It takes the shipped numbers at face value, drives them through the same summation the app performs, and compares against the source figures. A fit that agrees with itself proves nothing; this closes the loop. It also contains a replica of the previous code path so the defect magnitudes in Section 7 are measured rather than asserted.

Keeping the replica in sync. If you change a constant in `PharmacokineticConstants.swift`, change it in `pk_engine_replica.py` too. Every validation script reads from the replica, so a divergence silently invalidates all of them.

11 Sources

Bi Y. et al. (2018)
"Population Pharmacokinetic/Pharmacodynamic Modeling of Depot Testosterone Cypionate in Healthy Male Subjects." *CPT: Pharmacometrics & Systems Pharmacology.* `doi:10.1002/psp4.12287`.

Primary. Table 2 supplies CL/F, V/F and K_a with RSEs and bootstrap CIs; Figure 2 supplies the out-of-sample steady-state check. The only primary population-PK source used anywhere in the model.

AndroGel 1% prescribing information
FDA label, §12.3 and Table 5.
Regulatory primary. Supplies transdermal $F = 0.10$ verbatim, the Day-180 steady-state C_{avg} by dose arm, and the washout statement used for the terminal half-life. AndroGel 1.62% Table 6 supplies the corroborating dose-response inversion discussed in Section 8.

Swerdloff R.S. et al. (2020) / Jatenzo label
Oral testosterone undecanoate.
Regulatory primary. Supplies C_{avg}, $C_{\max}$ and the $t_{\max}$ window at 237 mg BID, with the SD used to argue mean-of-maxima bias.

transfemscience.org compiled tables (9 and 10)
Estradiol ester PK.
Secondary, high quality. A compilation of published single-dose and steady-state estradiol data. Used for EV and EC. Not a primary table, and flagged as such.

Wang C. et al. (2010)
Injectable testosterone undecanoate.
Secondary. Cited for the ∼1100 ng/dL single-dose $C_{\max}$ plausibility check on TU. Not fit to.

Endocrine Society clinical practice guidelines
Referenced for the therapeutic reference bands used to score the personas. Not used in any fit.

Source-quality summary. One primary population-PK paper (TC), two regulatory labels (T gel, oral T), one high-quality secondary compilation (EV, EC), and *nothing at all* for TE, TP and TU beyond the clearance identity that ties them to TC. Table 6 is the honest picture of what that buys: correct magnitudes everywhere, correct shape for four of the eight forms with a $t_{\max}$.

This document describes a consumer tracking application's internal model. It is not clinical guidance, and its predictions are population estimates that should not be used to make dosing decisions without laboratory confirmation.